

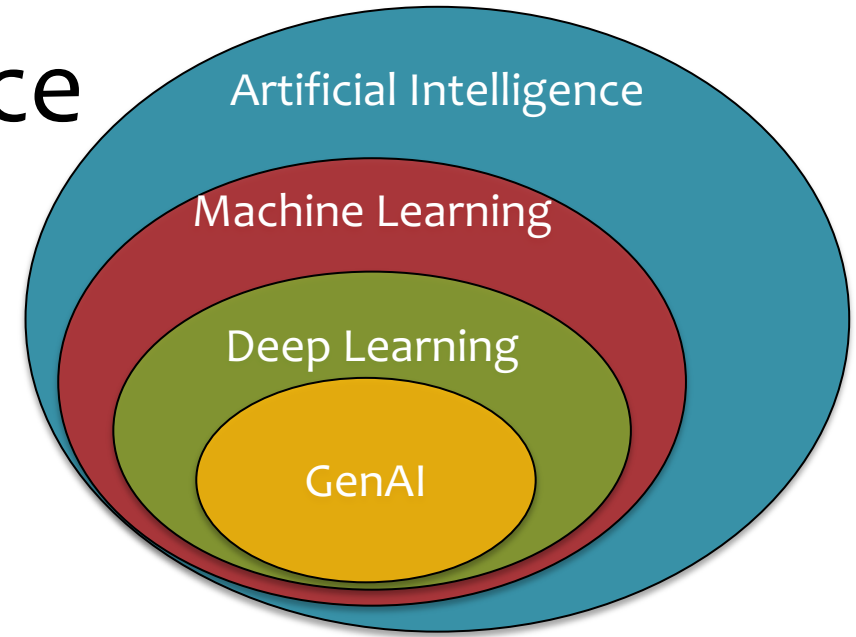
WHAT IS GENERATIVE AI?

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

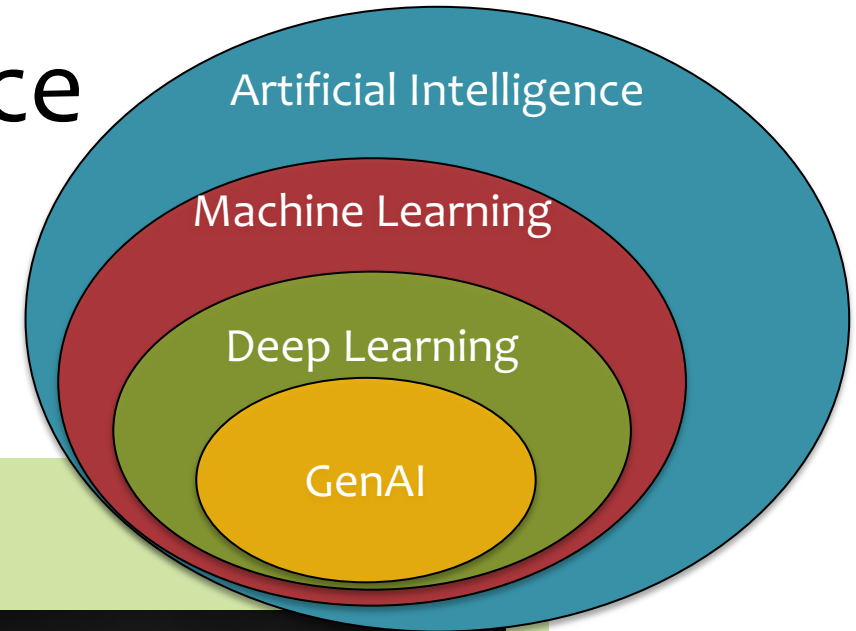


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

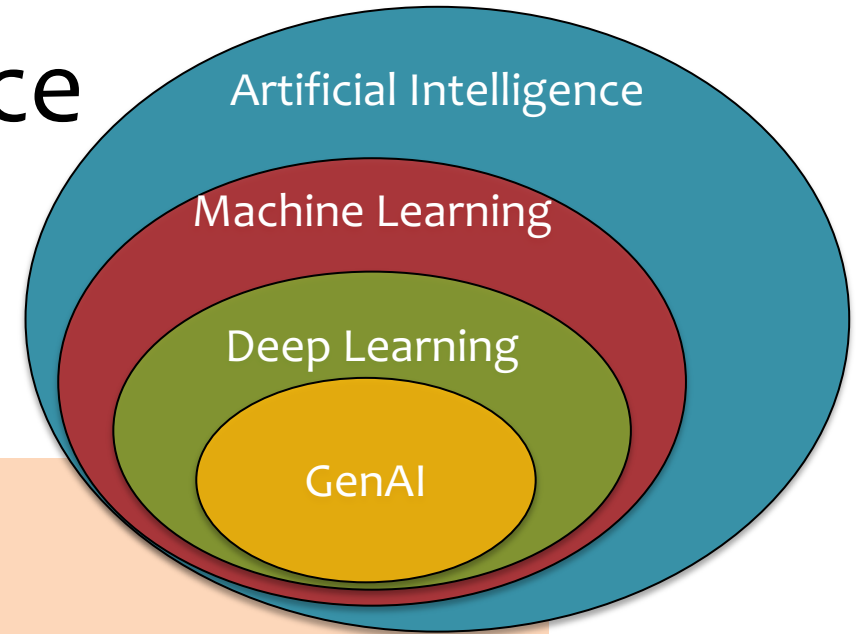


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

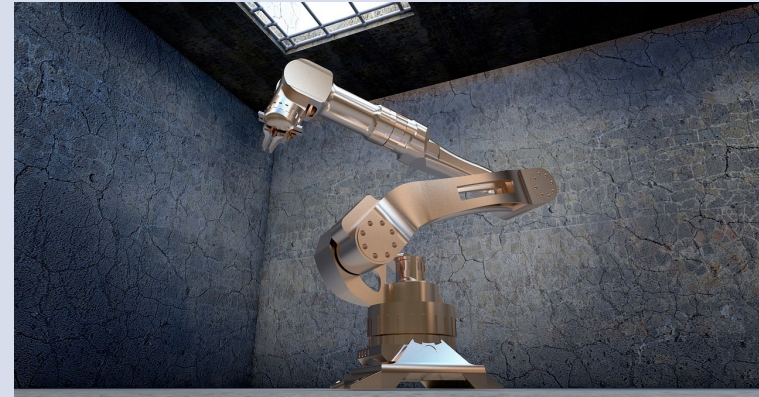
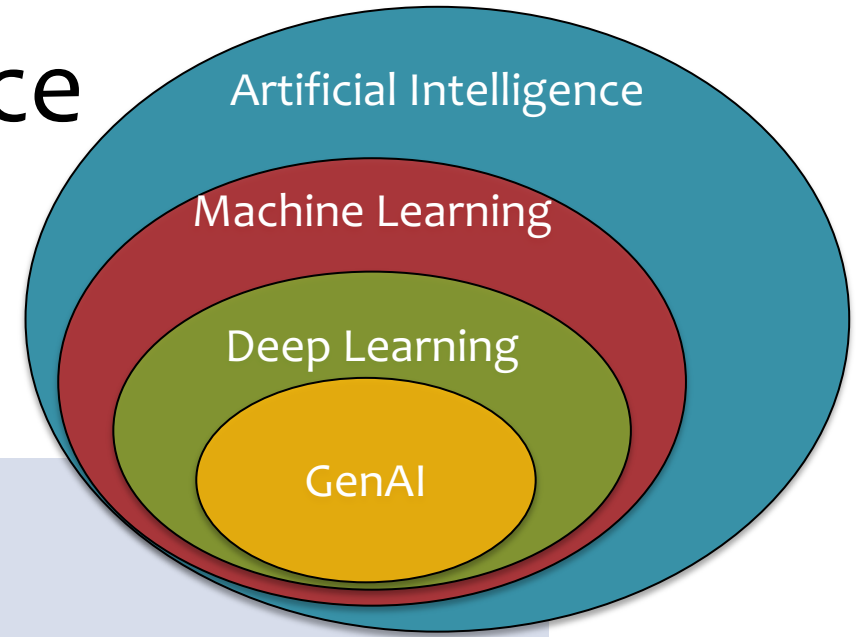


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

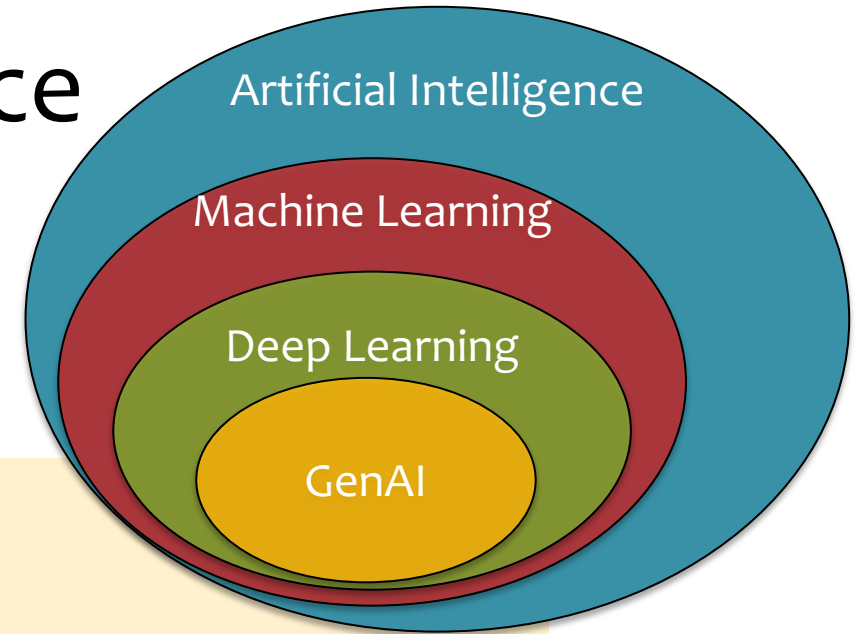


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

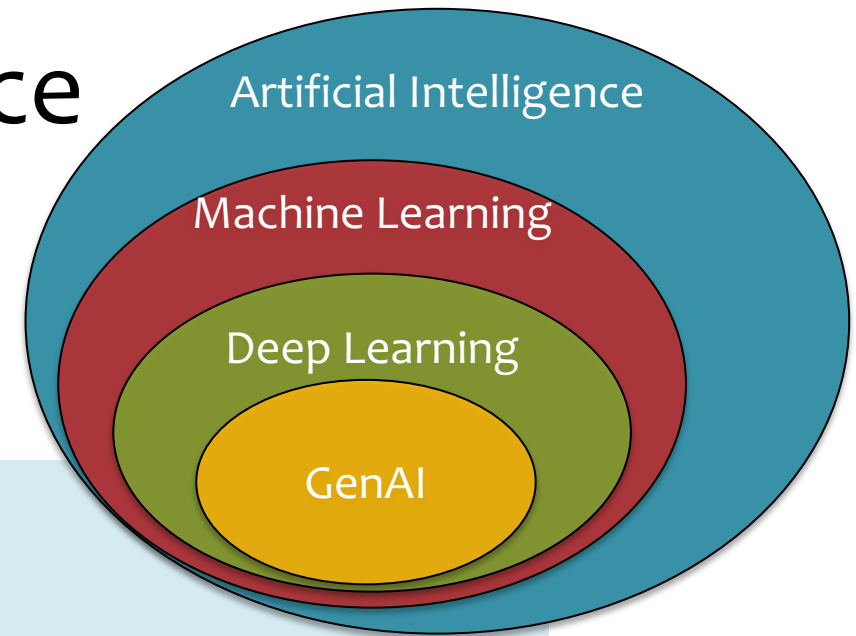


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

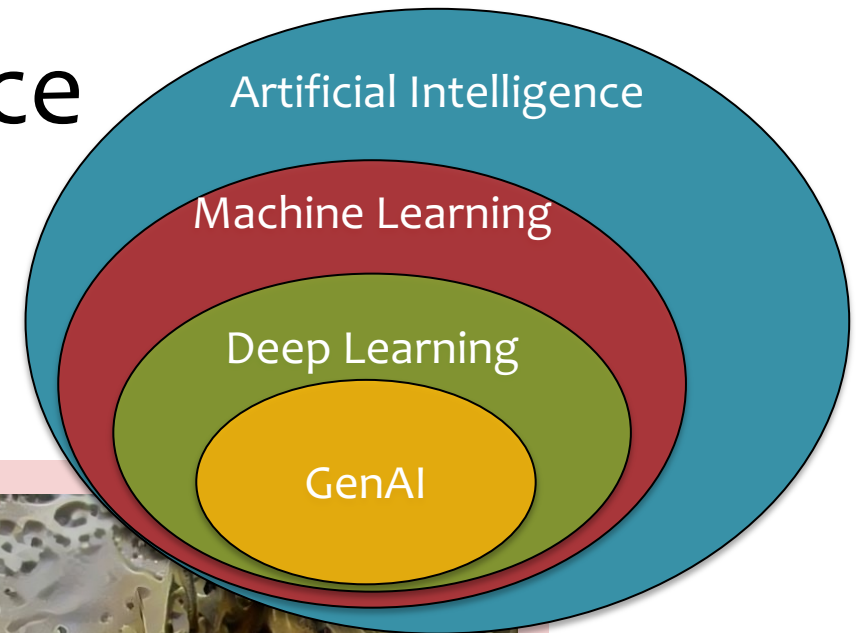


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

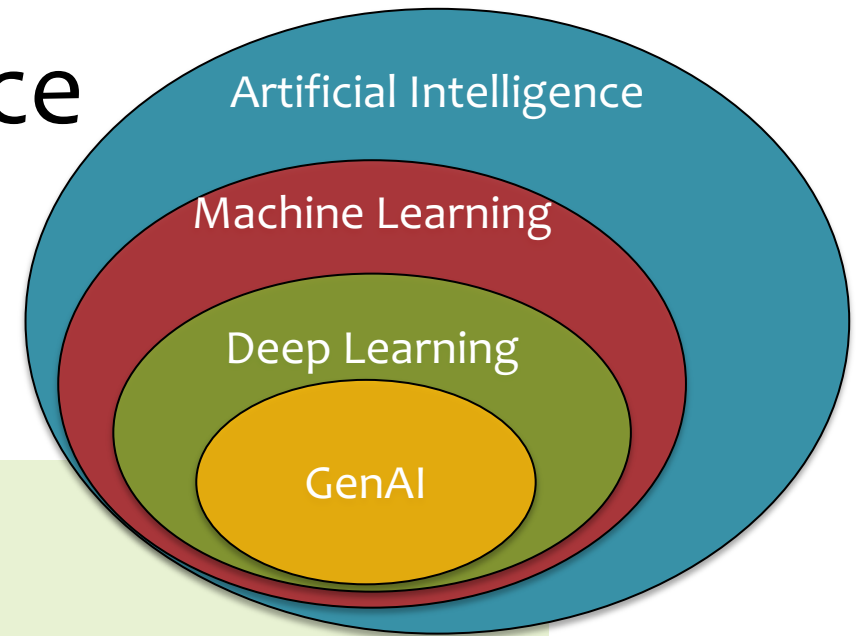


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning

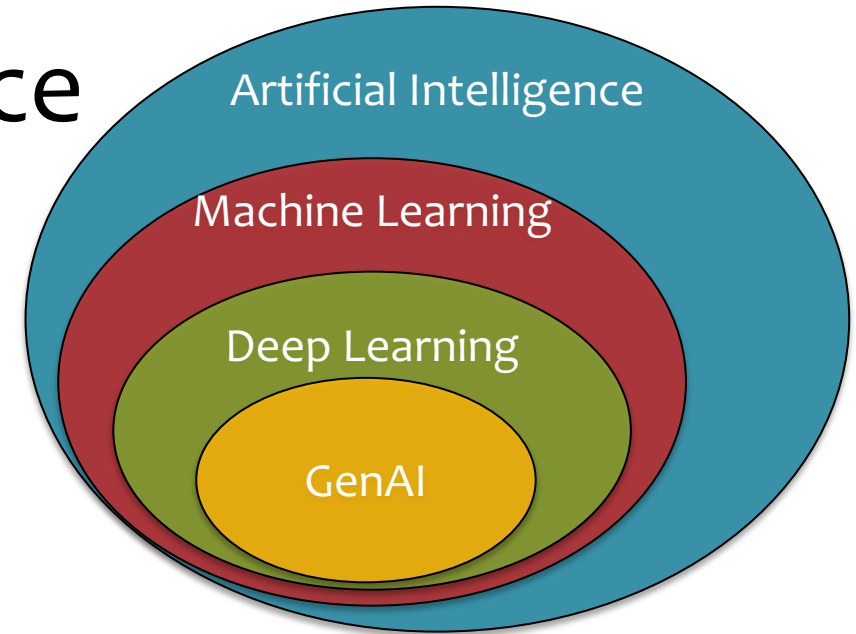


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



☐ Q: What does Generative AI have to do with **any of these goals?**

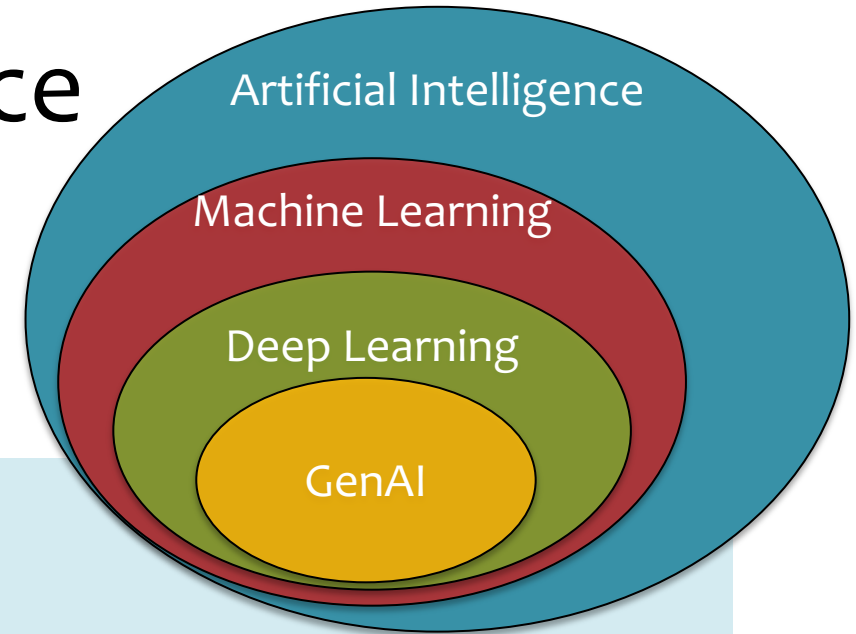
☐ A: It's making in-roads into **all of them.**

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



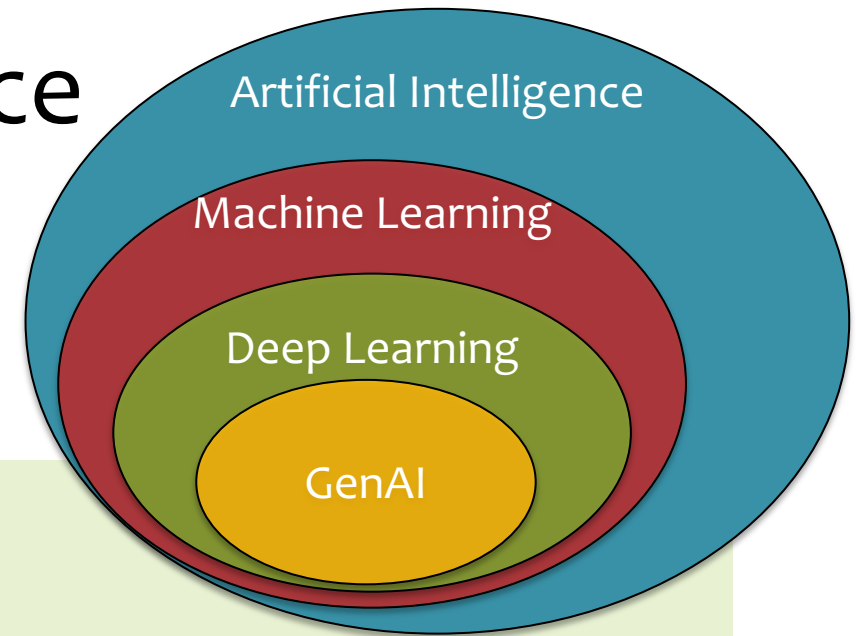
- ☐ Communication comprises the comprehension and generation of human language.
- ☐ Large language models (LLMs) excel at both
- ☐ (Even though they are most often trained autoregressively, i.e. to **generate** a next word, given the previous ones)

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



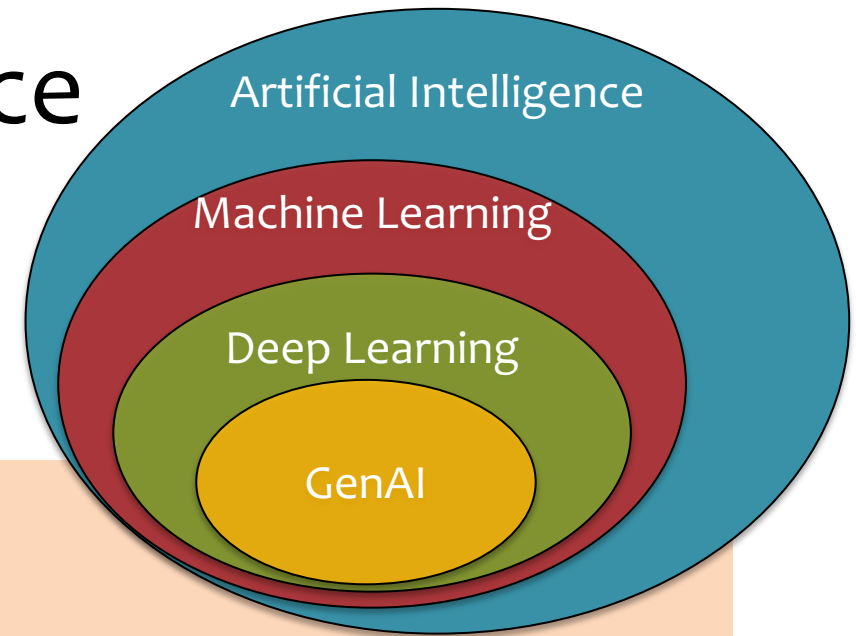
- ❑ The traditional way of learning in ML is via **parameter estimation**
- ❑ But **in-context learning** (i.e. providing training examples as context at test time) shows that learning can also be done via **inference**

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



- ❑ LLMs are also (unexpectedly) good at certain reasoning tasks
- ❑ cf. Chain-of-Thought Prompting (an ex. of in-context learning)

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

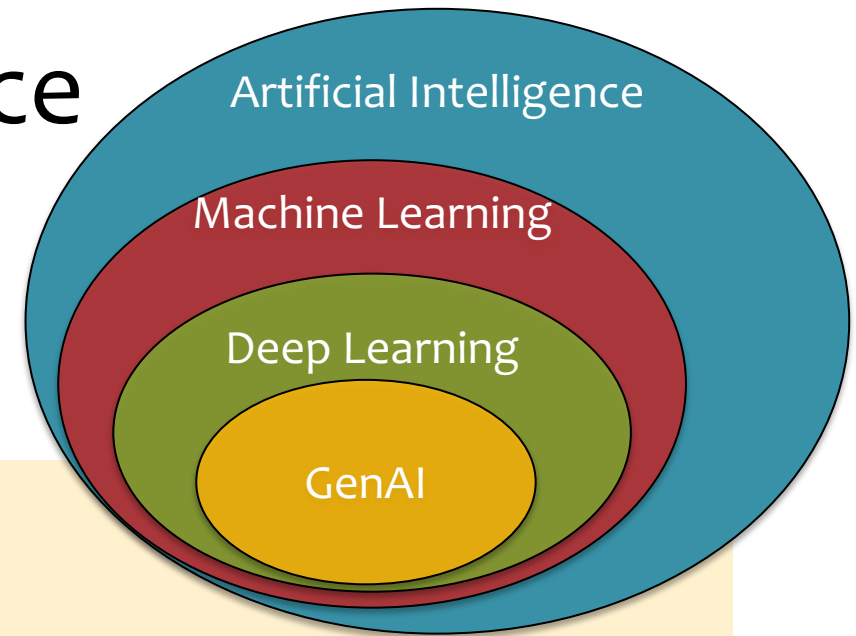
A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



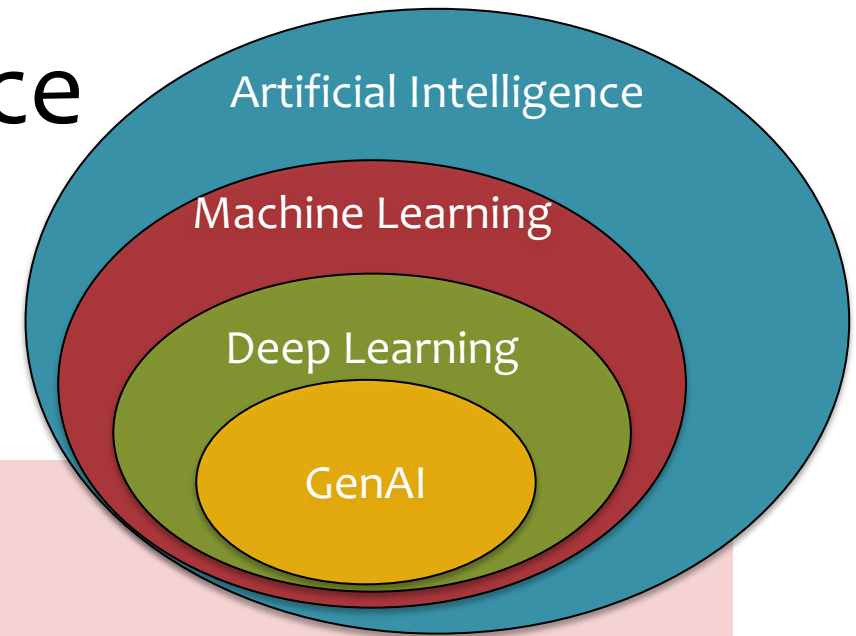
❑ LLMs are already being used for grounded planning for embodied agents, c.f. LLM-Planner

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



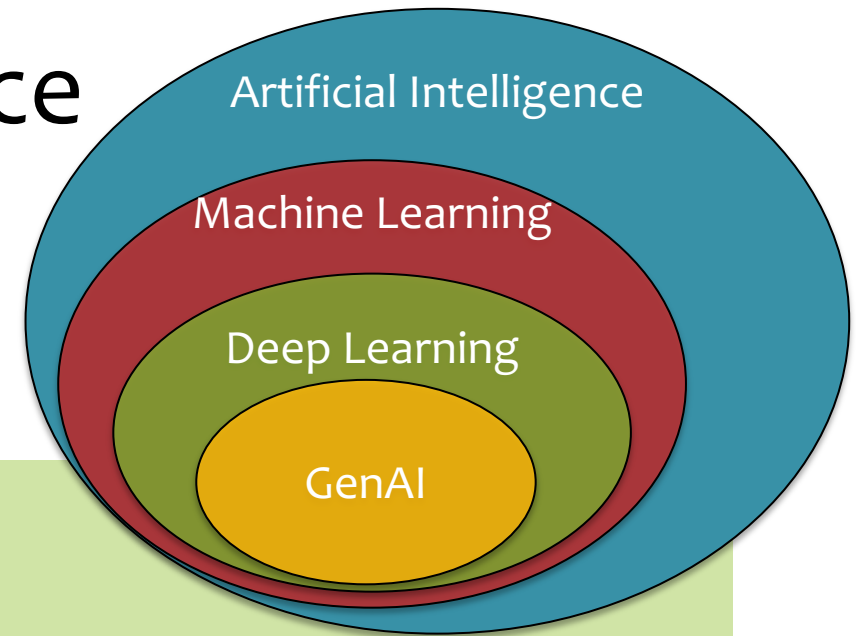
- ❑ Text-to-image models
[Midjourney's Discord server has 18 million members (1.7 million were online this morning)]
- ❑ Text-to-music models
[MusicGen capable of conditioning on text and audio sample]

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



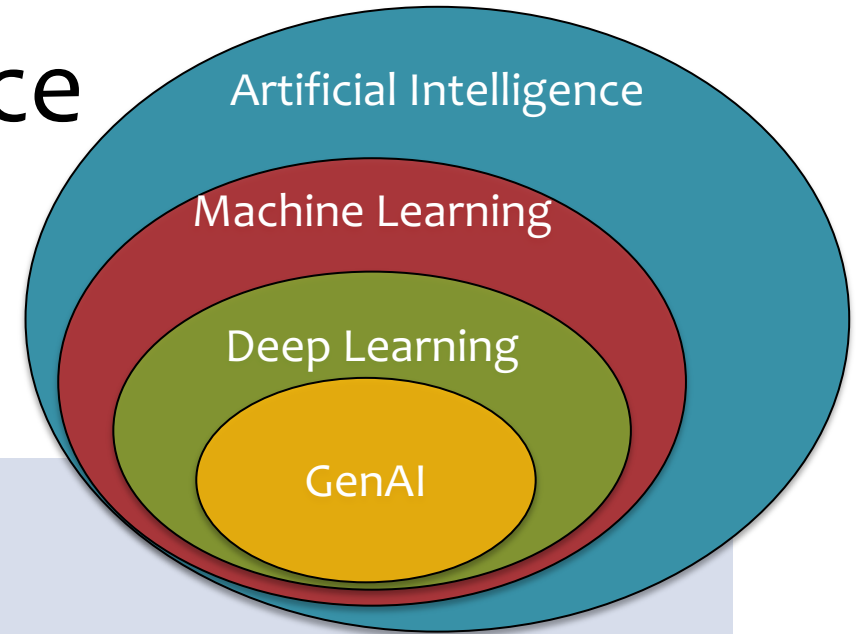
- ❑ Multimodal foundation models learn to answer questions about images (and text in images)
- ❑ Diffusion models can be used as zero-shot classifiers

Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

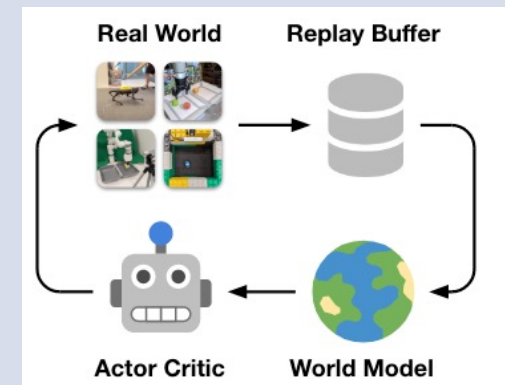
This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



❑ DayDreamer learns a generative model of experiences for RL, i.e. a World Model, without simulation

❑ Quadruped robot learns to walk in under 1 hour

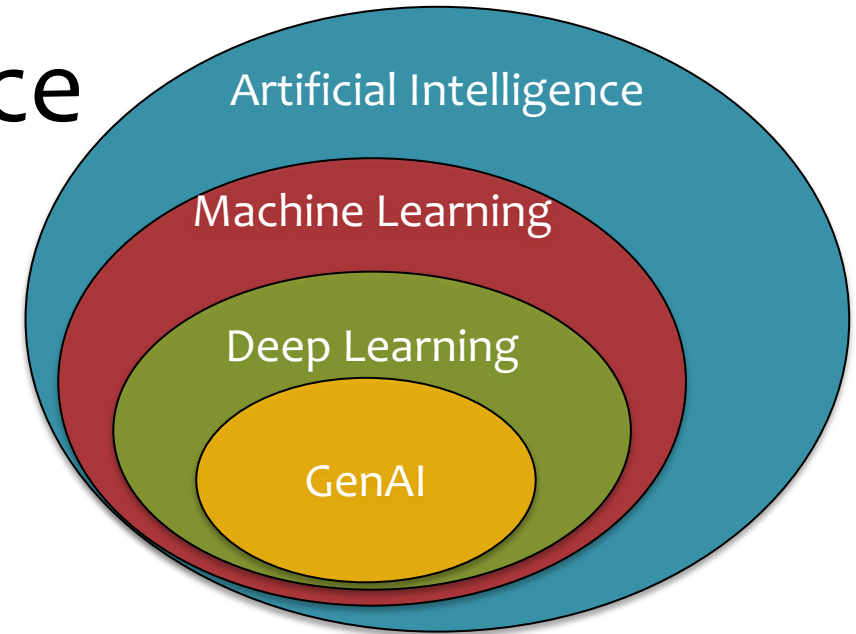


Artificial Intelligence

The basic goal of AI is to develop intelligent machines.

This consists of many sub-goals:

- Perception
- Reasoning
- Control / Motion / Manipulation
- Planning
- Communication
- Creativity
- Learning



❑ Q: What does Generative AI have to do with **any of these goals?**

❑ A: It's making in-roads into **all of them.**

WHAT IS GENERATIVE AI CAPABLE OF?

Text Generation

Prompt: Write a proof of the fact that there are infinitely many primes; do it in the style of a Shakespeare play through a dialogue between two parties arguing over the proof.

GPT-4: Enter FIRST PARTY and SECOND PARTY.

FIRST PARTY: Thou dost not understand, my friend,
There simply is no upper end To primes, those
numbers indivisible, No matter if it seems impossible.

SECOND PARTY: But how dost thou so surely know?
To prove this truth, what canst thou show?

FIRST PARTY: Consider this, my doubtful peer, A
clever proof that will make clear: Assume that there's
a final prime, The largest one we'll see in time. Let's
call this number "P" for short, And then, my friend,
we will purport To build a number "Q" that's new,
One larger than "P" and prime too.

SECOND PARTY: But how, my friend, will this be
done? The proof is not yet fairly won.

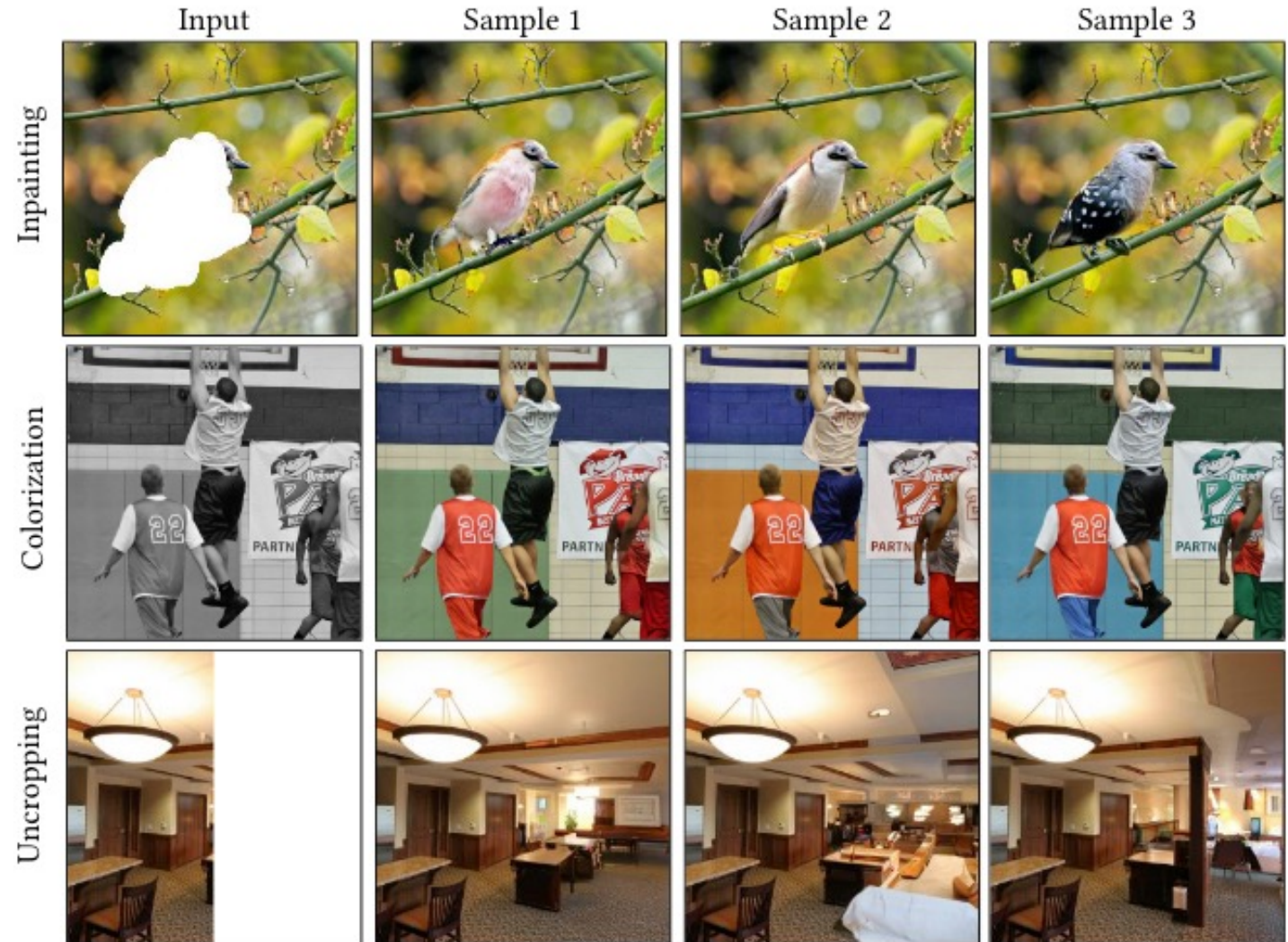
FIRST PARTY: It's simple, friend, just wait and see,
We'll multiply the primes, that's key. All primes from
two to "P" combined, Will make this new "Q" we will
find. But one thing more we must include, To make
the proof more watertight, We'll add a one onto the
end, And now the proof is in our sight.

SECOND PARTY: I see it now, it's crystal clear, For "Q"
must be a prime, my dear. It can't divide by any prime,
All smaller than itself in time. And since we chose "P"
as the last, This proves that our assumption's past.
There is no final prime, we see, The numbers go on
endlessly.

Image Editing

A variety of tasks involve automatic editing of an image:

- **Inpainting** fills in the (pre-specified) missing pixels
- **Colorization** restores color to a greyscale image
- **Uncropping** creates a photo-realistic reconstruction of a missing side of an image



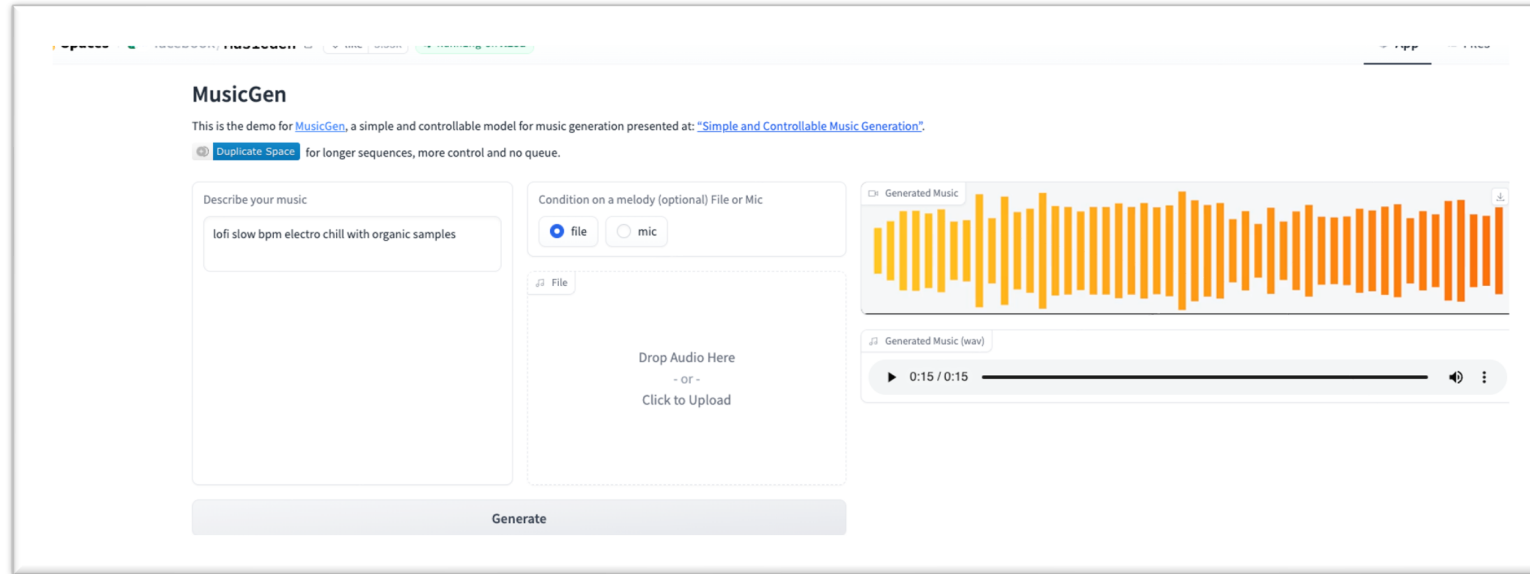
Text-to-Image Generation

- Given a text description, sample an image that depicts the prompt
- The following images are samples from SDXL with refinement

Prompt: close up headshot, futuristic **old man**, wild hair sly smile in front of gigantic UFO, dslr, sharp focus, dynamic composition, **rule of thirds**

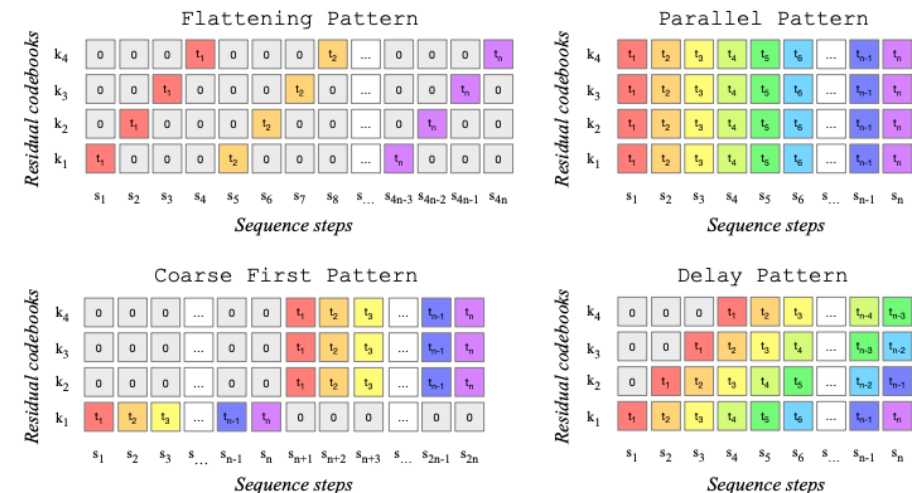


Music Generation



MusicGen

- A transformer decoder model over quantized units (discrete elements of a codebook of audio frames)
- Interleaves sounds by adjusting how codebooks attend to each other
- Permits conditioning on text and/or audio samples



Code Generation



Figure 1.3: We queried GPT-4 three times, at roughly equal time intervals over the span of a month while the system was being refined, with the prompt “Draw a unicorn in TikZ”. We can see a clear evolution in the sophistication of GPT-4’s drawings.

GPT-4 vs ChatGPT

Prompt:

A function f is called *increasing* if $f(s) \leq f(t)$ holds for every $s \leq t$. Consider the formula:
 $\lim_{x \rightarrow 0} \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n (f(nx) - \int_0^x \sin(nx) dx), \frac{(x + \frac{2}{8})^2 - 5}{4x - 1/x^2}, \dots, \text{partial } f(x/n) \right]$
 over partial $x * (e^{-i\omega n x} + 3(x \# 7))$, f inverse of $(x \bmod n)$]
 multiply $\text{matrix}[1, 1, \dots, 1 \setminus 1, 1, \dots, 1]$
 {grey_color{this gives us a good impression about the function}}

Convert the above paragraph to LaTeX.

GPT-4:

A function f is called *increasing* if $f(s) \leq f(t)$ holds for every $s \leq t$. Consider the formula:

$$\lim_{x \rightarrow 0} \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n (f(nx) - \int_0^x \sin(nx) dx), \frac{(x + \frac{2}{8})^2 - 5}{4x - 1/x^2}, \dots, \text{partial } f(x/n) \right] \times \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \end{bmatrix}$$

this gives us a good impression about the function

GPT-4 vs ChatGPT

Prompt:

I want to write a customized optimizer in pytorch, the optimizer should do:
 1. the parameters for the optimizer is learning rate, momentum, weight decay, d_dim, k and factor alpha
 2. the optimizer goes through each parameter in the network, and then
 -- Get the gradient of this parameter, reshape it to 2D by looping through the dimensions, and keeping the dimension whose actual dimension (shape) is equal to d_dim, do an SVD decomposition of the parameter so $W = U \Sigma V$.
 -- Keep the top k singular vectors of W , so we get a new matrix $W_k = U \Sigma_k V$.
 -- Keep the top 2k singular vectors of W , so we get a new matrix $W_{2k} = U \Sigma_{2k} V$.
 -- Normalize W_k according to the momentum of the F-norm of W_{2k} .
 -- Truncate each coordinate of W_k to the top alpha percentile of all the coordinates, in absolute value.
 -- Apply momentum on W_k , and then update the network parameters using this momentum.
 -- Apply weight decay.

GPT-4:

```
def step(self):
    for group in self.param_groups:
        for p in group['params']:
            if p.grad is None:
                continue
            grad = p.grad.data
            # reshape grad to 2D
            grad_2d = grad.reshape(-1, group['d_dim'])
            # SVD decomposition
            U, S, V = torch.svd(grad_2d)
            # keep top k singular vectors
            G_k = U[:, :group['k']] @ torch.diag(S[:group['k']]) @ V[:, :group['k']].T
            # keep top 2k singular vectors
            G_2k = U[:, :2*group['k']] @ torch.diag(S[:2*group['k']]) @ V[:, :2*group['k']].T
            # normalize G_k
            G_k /= torch.norm(G_2k, p=2)
            # truncate G_k
            threshold = torch.quantile(torch.abs(G_k),
                                     group['alpha'])
            G_k[torch.abs(G_k) < threshold] = 0
            # apply momentum
            if 'momentum_buffer' not in group:
                group['momentum_buffer'] = torch.zeros_like(G_k)
            group['momentum_buffer'] = group['momentum_buffer'] +
            group['momentum_buffer'] + G_k
            # update parameters
            p.data.add_(-group['lr'], group['momentum_buffer'])
            # apply weight decay
            if group['weight_decay'] != 0:
                p.data.add_(-group['weight_decay'] * group['lr'],
                           p.data)
```

ChatGPT:

```
def step(self):
    for group in self.param_groups:
        for p in group['params']:
            if p.grad is None:
                continue
            grad = p.grad.data
            grad = grad.reshape(-1, grad.shape[-1]) # reshape to 2D
            # SVD decomposition
            u, s, v = torch.svd(grad)
            # Keep top k and 2k singular vectors
            k = group['k']
            u_k = u[:, :k]
            s_k = s[:k]
            v_k = v[:, :k]
            u_2k = u[:, :2*k]
            s_2k = s[:2*k]
            v_2k = v[:, :2*k]
            # Normalize according to momentum of F-norm of G_2k
            grad_k = u_k @ torch.diag(s_k) @ v_k.T()
            norm_2k = s_2k.norm()
            grad_k = grad_k * group['momentum'] / norm_2k
            # Truncate to top alpha percentile
            alpha = group['alpha']
            percentile = grad_k.abs().flatten().kthvalue(int(alpha *
            grad_k.numel()))
            grad_k[grad_k.abs() < percentile] = 0
            # Apply weight decay
            p.data.add_(-group['lr'], grad_k + group['weight_decay']
            * p.data)
```

Figure 3.4: GPT-4 vs ChatGPT at implementing a customized Pytorch optimizer module. We highlight the **astuteness** of GPT-4 vs the **mistake** by ChatGPT at ‘applying momentum’.

Video Generation

- Latent diffusion models use a low-dimensional latent space for efficiency
- **Key question:** how to generate multiple correlated frames?
- ‘Align your latents’ inserts temporal convolution / attention between each spatial convolution / attention
- ‘Preserve Your Own Correlation’ includes temporally correlated noise



SCALING UP

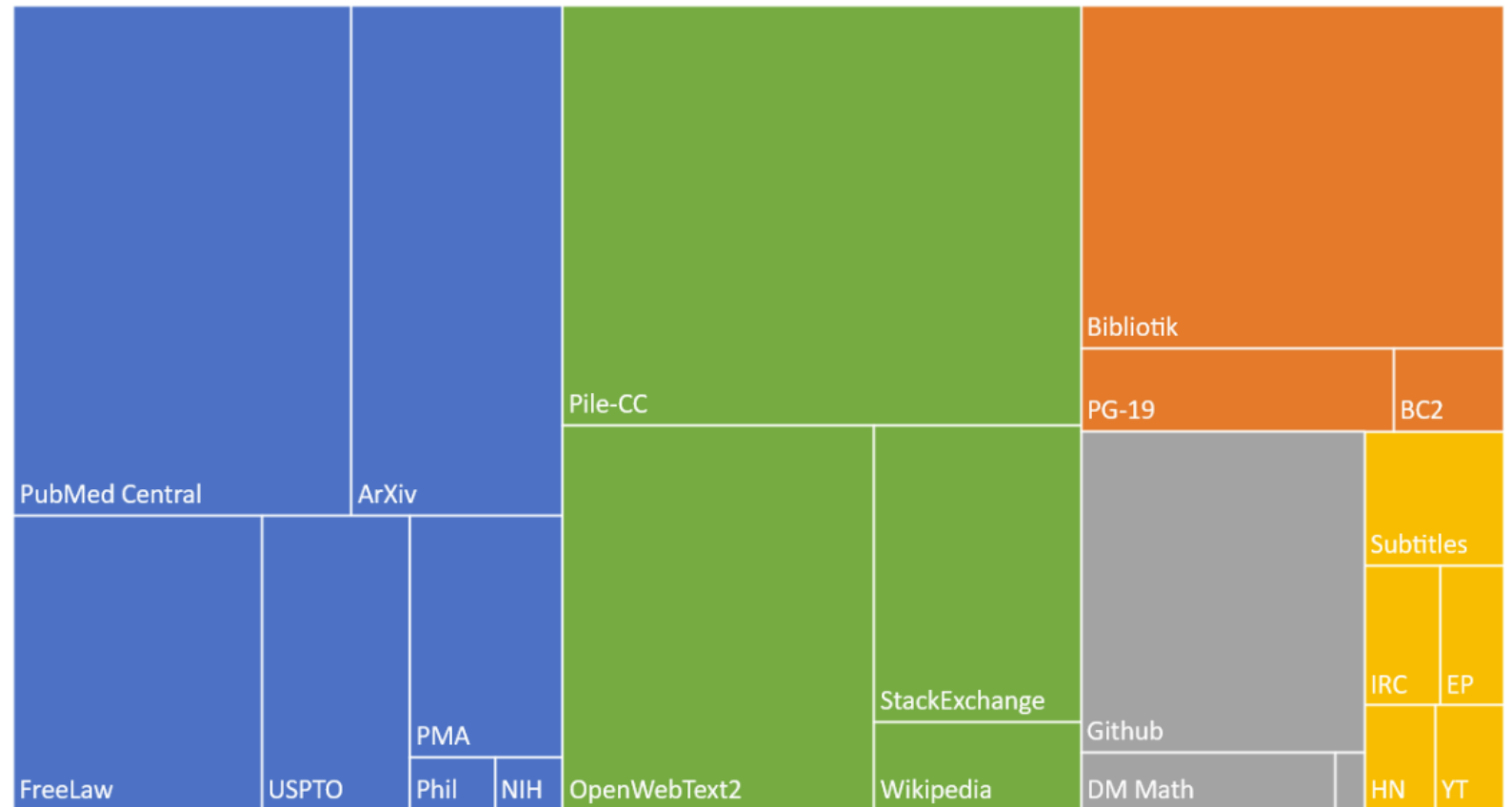
Training Data for LLMs

The Pile:

- An open source dataset for training language models
- Comprised of 22 smaller datasets
- Favors high quality text
- 825 Gb $\approx$ 1.2 trillion tokens

Composition of the Pile by Category

■ Academic ■ Internet ■ Prose ■ Dialogue ■ Misc



RLHF

- **InstructGPT** uses Reinforcement Learning with Human Feedback (RLHF) to **fine-tune a pre-trained GPT model**
- From the paper: “In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters.”

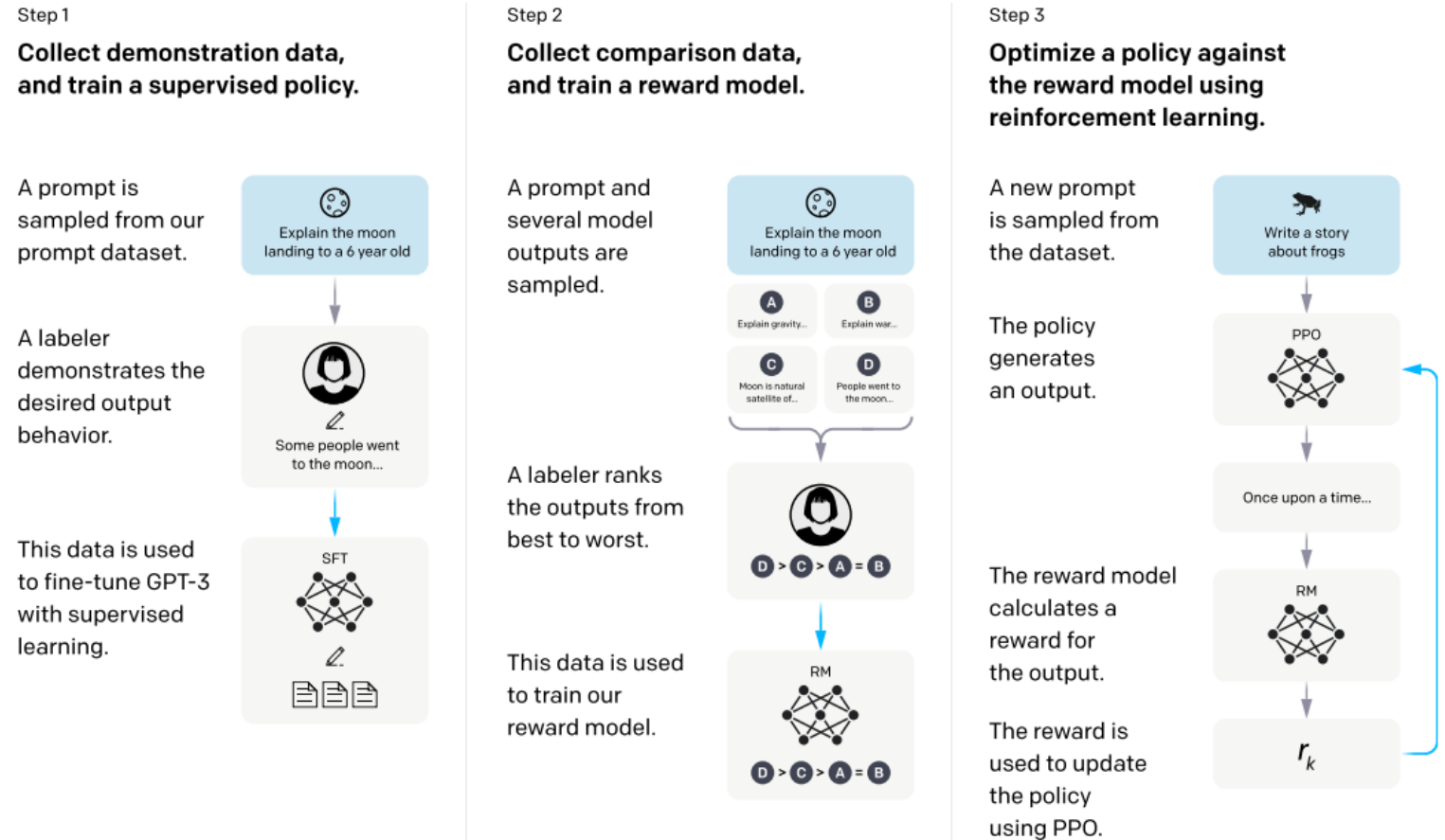


Figure 2: A diagram illustrating the three steps of our method: (1) supervised fine-tuning (SFT), (2) reward model (RM) training, and (3) reinforcement learning via proximal policy optimization (PPO) on this reward model. Blue arrows indicate that this data is used to train one of our models. In Step 2, boxes A-D are samples from our models that get ranked by labelers. See Section 3 for more details on our method.

Memory Usage of LLMs

How to store a large language model in memory?

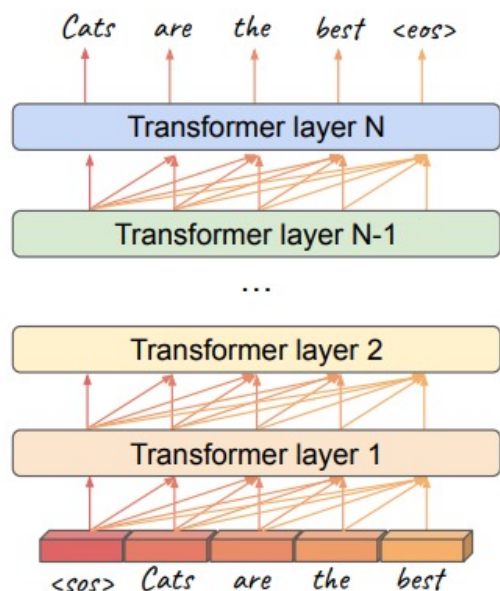
- **full precision:** 32-bit floats
- **half precision:** 16-bit floats
- Using half precision not only **reduces memory**, it also **speeds up** GPU computation
- “*Peak float16 matrix multiplication and convolution performance is 16x faster than peak float32 performance on A100 GPUs.*”

[from Pytorch docs](#)

Model	Megatron-LM	GPT-3
# parameters	8.3 billion	175 billion
full precision	30 Gb	651 Gb
half precision	15 Gb	325 Gb

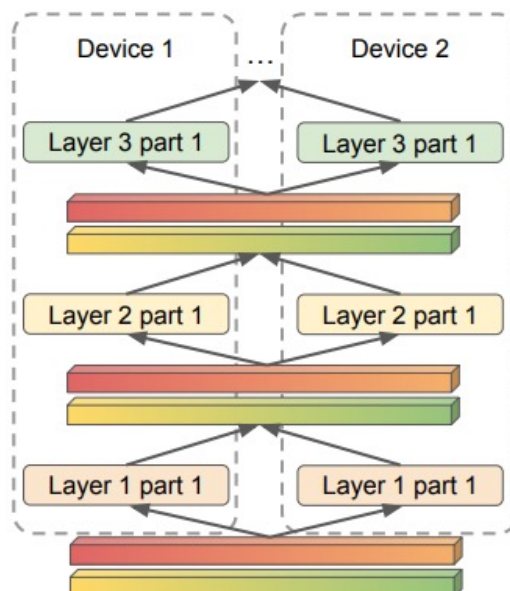
GPU / TPU	Max Memory
TPU v2	16 Gb
TPU v3/v4	32 Gb
Tesla V100 GPU	32 Gb
NVIDIA RTX A6000	48 Gb
Tesla A100 GPU	80 Gb

Distributed Training: Model Parallel



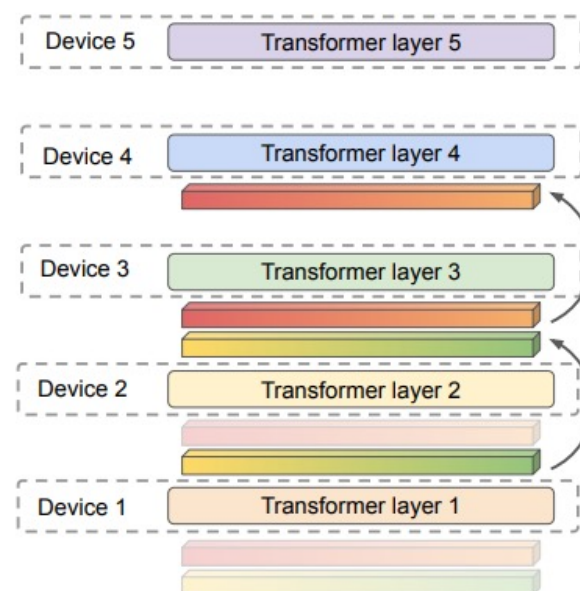
(a) Transformer-based LM

There are a variety of different options for how to distribute the model computation / parameters across multiple devices.



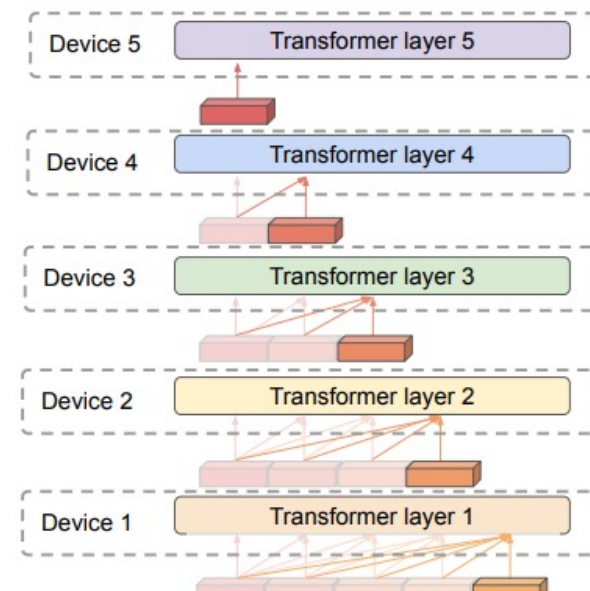
(b) Operation partitioning (Megatron-LM)

Matrix multiplication comprises most Transformer LM computation and can be divided along rows/columns of the respective matrices.



(c) Microbatch-based pipeline parallelism (GPipe)

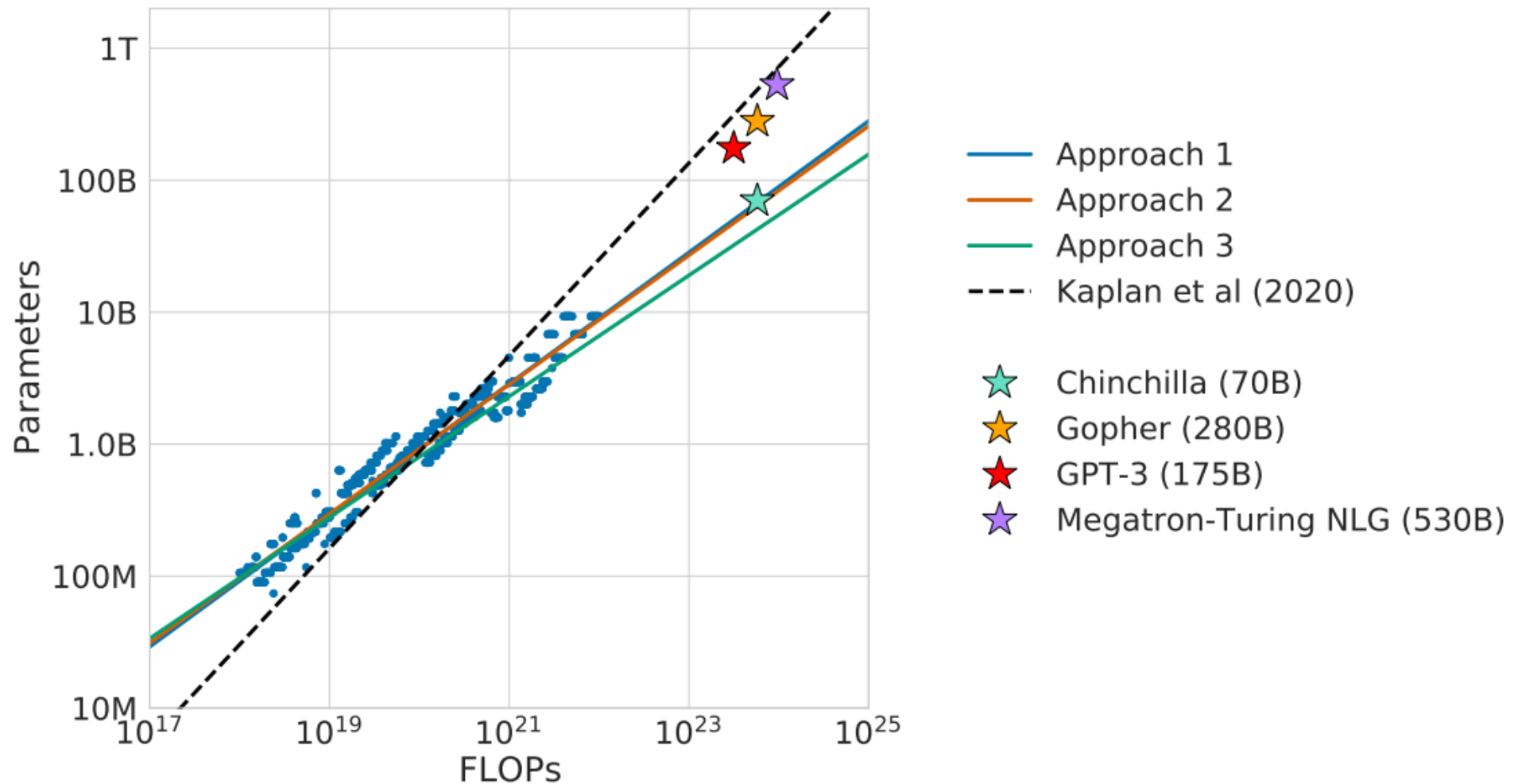
The most natural division is by layer: each device computes a subset of the layers, only that device stores the parameters and computation graph for those layers.



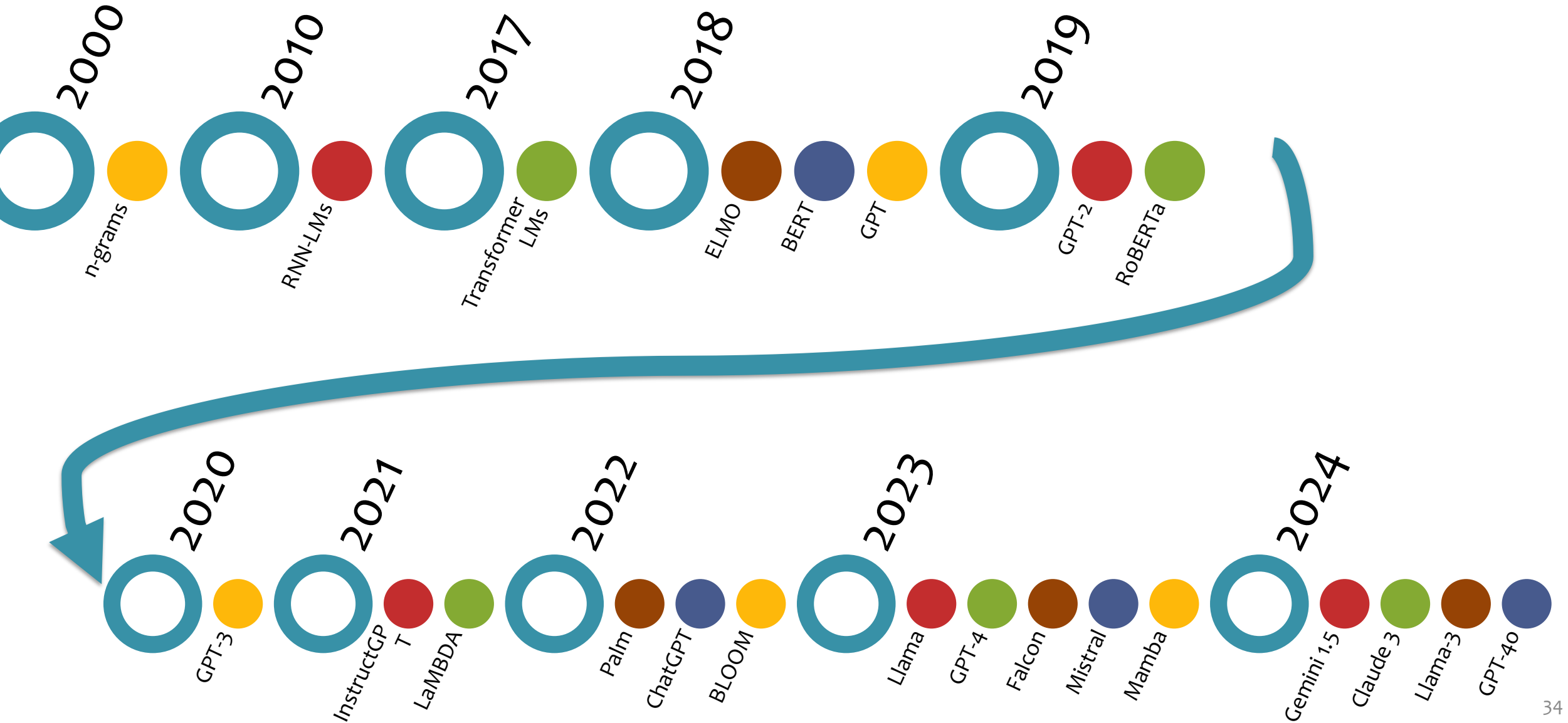
(d) Token-based pipeline parallelism (TeraPipe)

A more efficient solution is to divide computation by token *and* layer. This requires careful division of work and is specific to the Transformer LM.

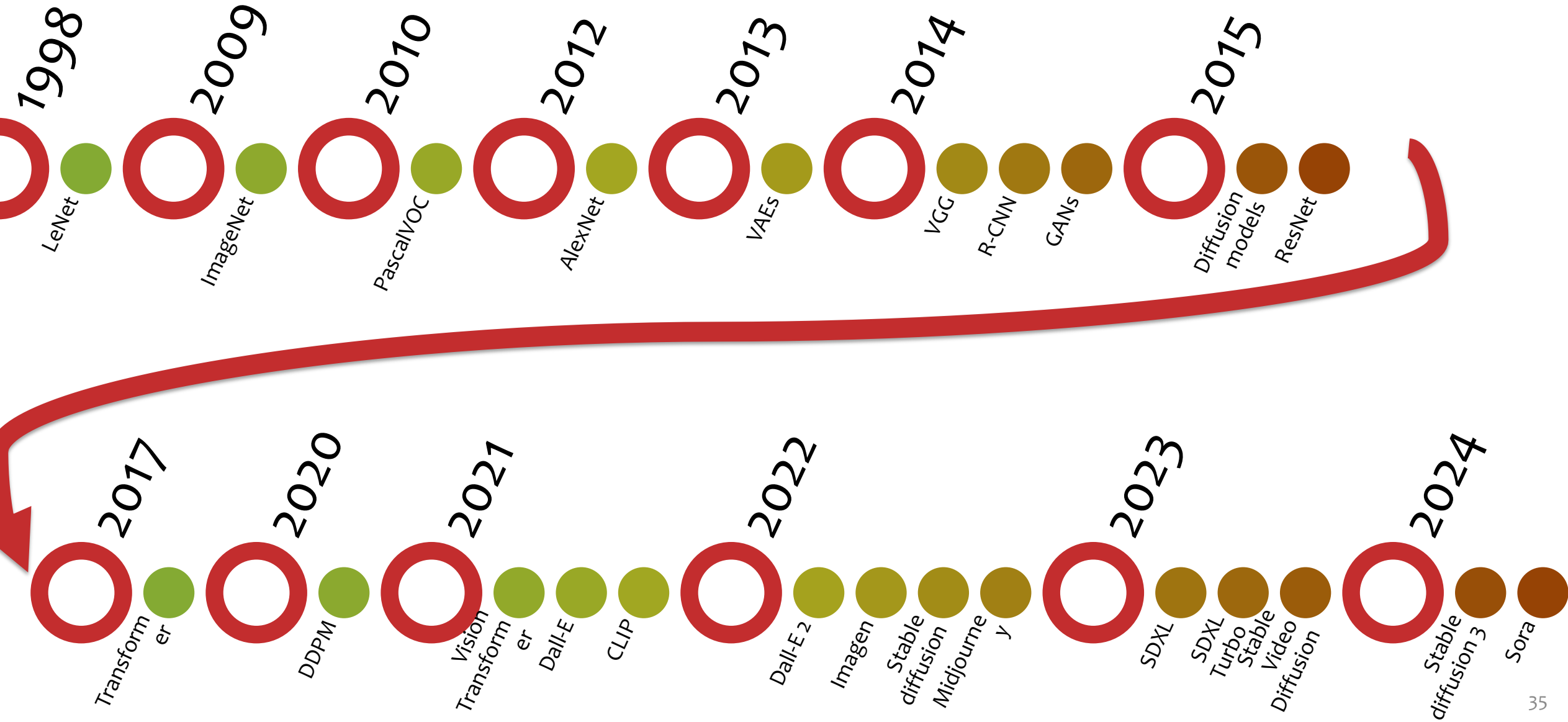
Cost to train



Timeline: Language Modeling



Timeline: Image Generation



**Why learn the inner-
workings of GenAI?**

(a metaphor)



Figure from <https://www.astonmartin.com/en/>



Figure from <https://daily.jstor.org/the-science-of-traffic/>

2021 ties 2018 for Sixth Warmest Year on Record
Global Temperature Anomaly (°C compared to the 1951-1980 average)

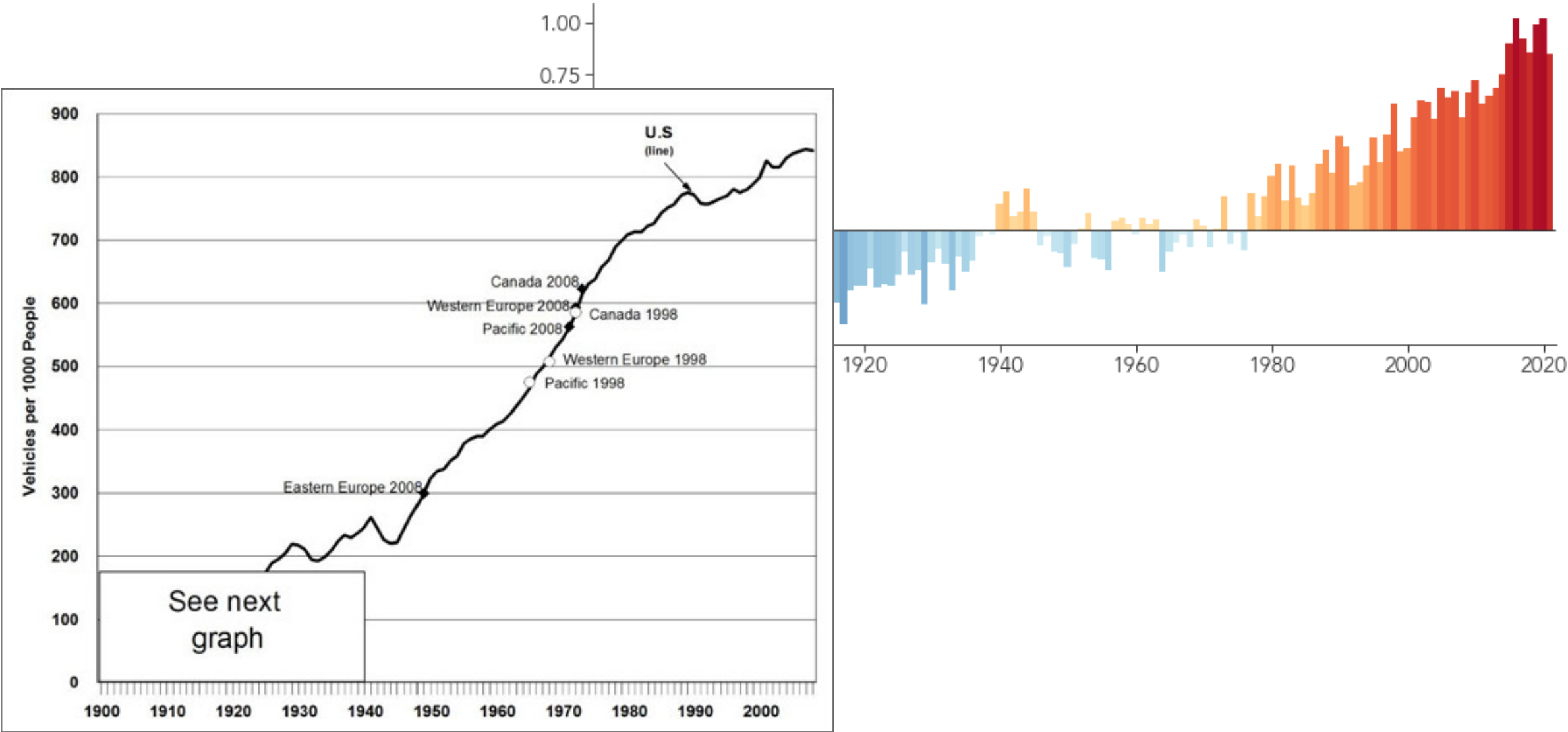
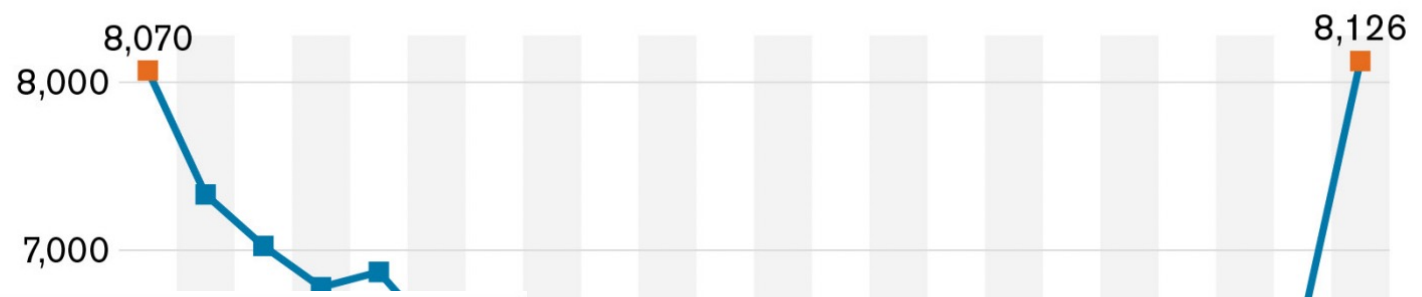
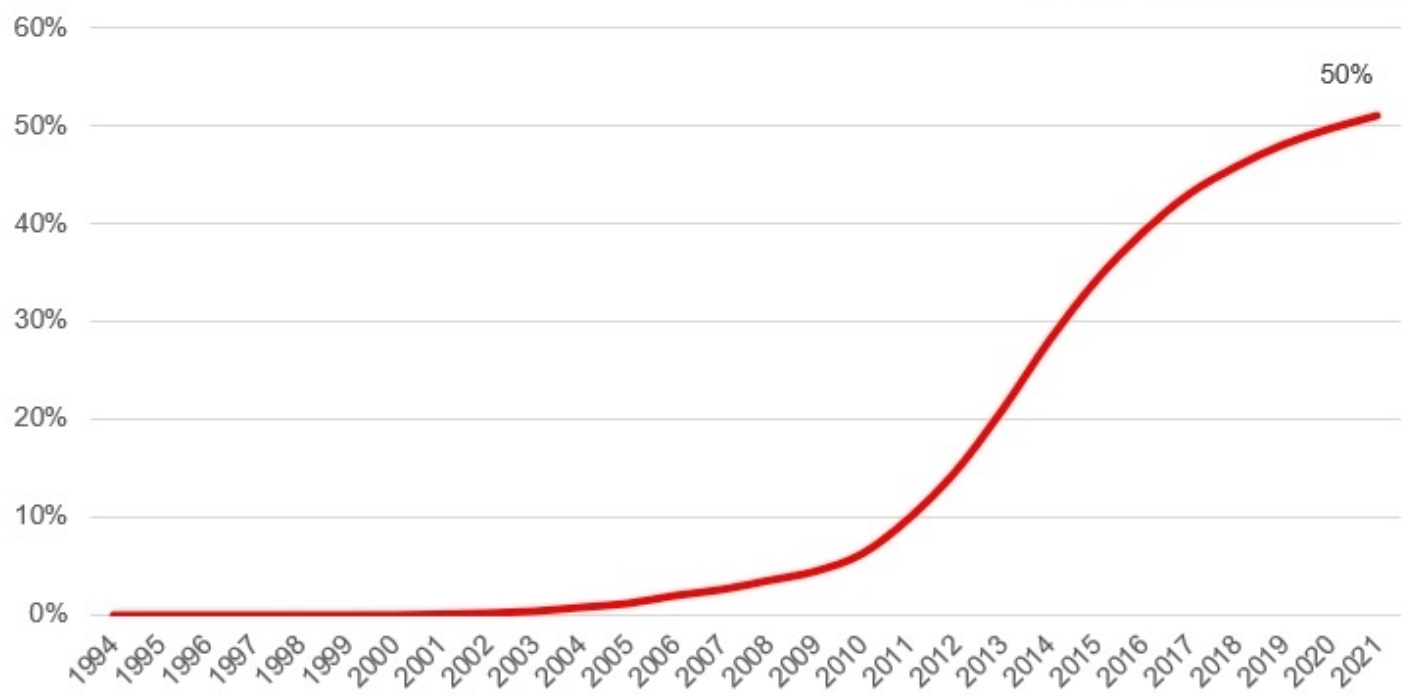


Figure from [GHSA](#)

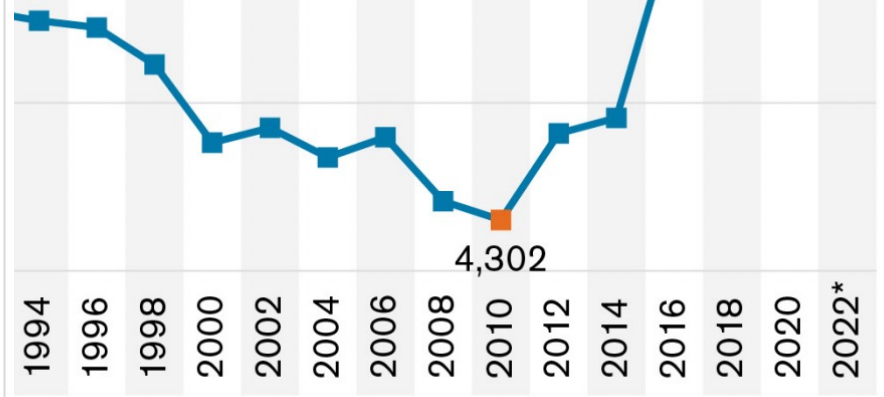
Figure 5 Number of Annual U.S. Pedestrian Fatalities, 1980-2022



% of World Population That Uses a Smartphone



STRATEGYANALYTICS





GENERATIVE AI IS PROBABILISTIC MODELING

GenAI is Probabilistic Modeling

$$p(x_{t+1} \mid x_1, \dots, x_t)$$

**What if I want to model
EVERY possible
interaction?**

**...or at least the interactions of the
current variable with all those that came
before it...**

(RNN-LMs)

RNN Language Model

$$\text{RNN Language Model: } p(w_1, w_2, \dots, w_T) = \prod_{t=1}^T p(w_t \mid f_{\theta}(w_{t-1}, \dots, w_1))$$

$$p(w_1, w_2, w_3, \dots, w_6) =$$

The						$p(w_1)$
The	bat					$p(w_2 \mid f_{\theta}(w_1))$
The	bat	made				$p(w_3 \mid f_{\theta}(w_2, w_1))$
The	bat	made	noise			$p(w_4 \mid f_{\theta}(w_3, w_2, w_1))$
The	bat	made	noise	at		$p(w_5 \mid f_{\theta}(w_4, w_3, w_2, w_1))$
The	bat	made	noise	at	night	$p(w_6 \mid f_{\theta}(w_5, w_4, w_3, w_2, w_1))$

Key Idea:

- (1) convert all previous words to a **fixed length vector**
- (2) define distribution $p(w_t \mid f_{\theta}(w_{t-1}, \dots, w_1))$ that conditions on the vector

Topics

- Generative models of text
 - RNN LMs / Autodiff
 - Transformer LMs
 - Pre-training, fine-tuning, evaluation, decoding
- Generative models of images
 - CNNs / Transformers for vision
 - GANs, Conditional GANs
 - VAEs and Diffusion models
- Applying and adapting foundation models
 - Reinforcement learning with human feedback (RLHF)
 - Parameter-efficient fine tuning
 - In-context learning for text
 - In-context learning for vision
- Multimodal foundation models
 - Text-to-image generation
 - Aligning multimodal representations
 - Visual-language foundation models
- Scaling models
 - Efficient decoding strategies
 - Distributed training
 - Scaling laws and data
 - Mixture of experts / FlashAttention
- What can go wrong?
 - Safety/bias/fairness, Hallucinations, Adversarial (e.g., prompt injection) attacks
 - Cheating – how to watermark, Legal issues, e.g., copyright,...
 - Drift in performance, Data contamination, Lack of ground truth
- Advanced Topics
 - State space models
 - Code generation
 - Audio understanding and synthesis
 - Video synthesis